Course and its outcomes

II Semester: Quantum Mechanics – II (B010801T)

(Part B: Relativistic Quantum Mechanics)

Unit IV Klein Gordon Equation and Free Particle, Solution, Dirac Equation, Dirac Matrices, Covariance of Dirac Equation & Bilinear Covariants,

Unit V Solution for a Free Particle, Negative Energy states and Hole Theory, Spin, Position Operator.

Reading List:

1. Quantum Mechanics, L. I. Schiff, Mc-Graw Hill
2. Relativistic Quantum Mechanics, James D. Bjorken and Sidney D. Drell (Tata McGraw Hill Education, 2013)

Learning Outcomes

Knowledge:

The students

- Will be able to explain why Schrodinger equation fails at relativity.
- Will be able to analyze the Klein-Gordon and explain problems arising while dealing with it.
- Will be able to explain how Dirac addressed problems that occurred in the case of Klein Gordon equation.
- Will have fundamental concept of spin of particles.

Skills:

The students

- Will be able to derive Klein-Gordon equation and formulation for probability and current density.
- Will be able to perform solutions of Klein-Gordan and Dirac equations for free particle.